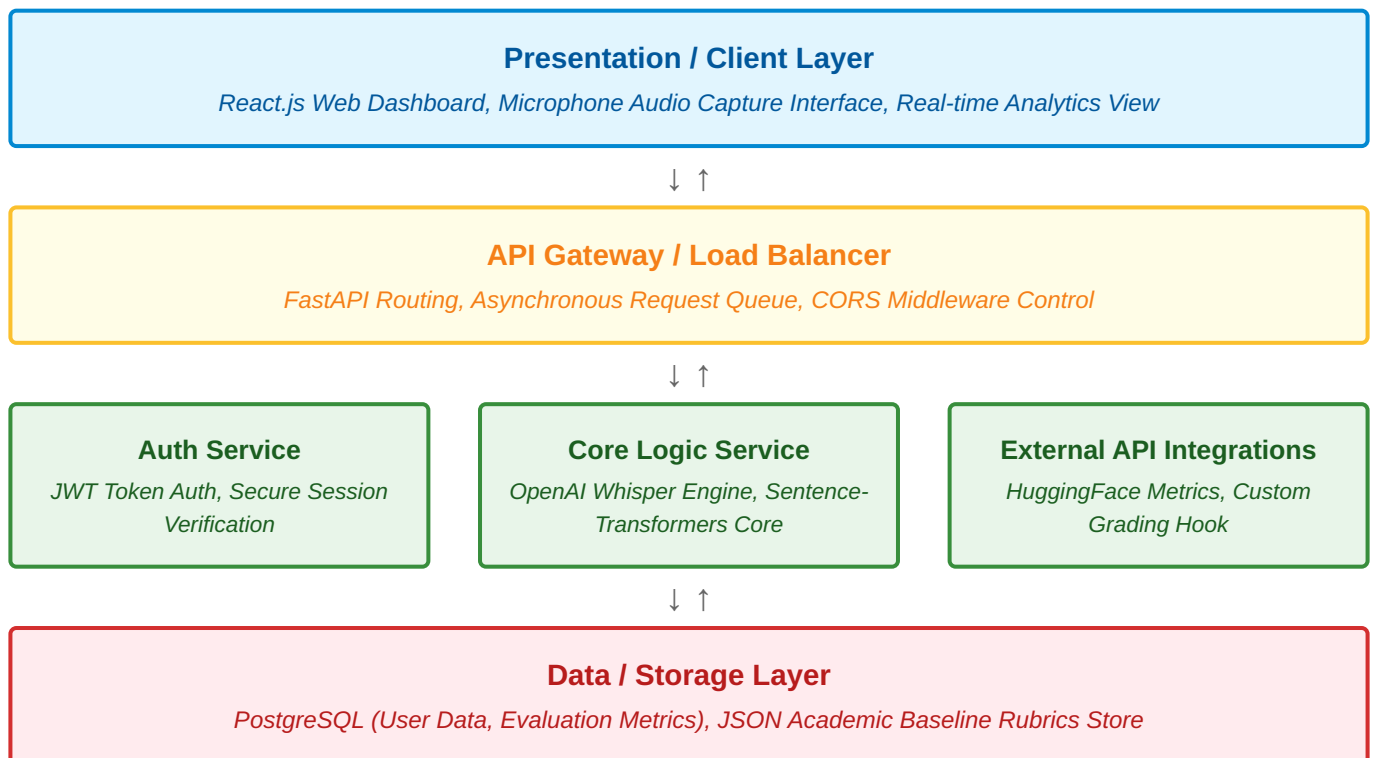


SOLUTION ARCHITECTURE

Date	3 July 2026
Team ID	
Project Name	Voice based concept understanding analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



COMPONENT DESCRIPTION TABLE

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the user workspace for students and reviewers. Captures real-time microphonic speech data stream chunks, handles authentication forms, and displays final concept evaluation metrics graphs.	React.js, Web Audio API, TailwindCSS
API Gateway	Acts as the single point of entry to securely intercept client requests, manage high-concurrency request distribution pipelines, check JWT credentials, and handle CORS routes smoothly.	FastAPI Uvicorn Core, Python Gateway Layers
Microservices	Processes raw acoustic sound inputs through automatic speech recognition systems, and validates vector semantic distances against target curriculum benchmarks to dynamically score understanding details.	OpenAI Whisper, Sentence-Transformers, scikit-learn
Database	Maintains permanent storage configurations for secure account parameters, individual assessment result transcripts, and structured key-phrase configuration lists mapped to academic grading rubrics.	PostgreSQL, JSON Local Blob Store